Suhrid: A Collaborative Mobile Phone Interface for Low Literate People

Syed Ishtiaque Ahmed[1], Maruf Hasan Zaber[2], Mehrab Bin Morshed[2], Md.Habibullah Bin Ismail[2], Dan Cosley[1], Steven J. Jackson[1]
[1]Department of Information Science, Cornell University, Ithaca, NY, USA 14850
[2]Department of Computer Science and Engineering, BUET, Dhaka, Bangladesh 1000
{sa738, drc44, sjj54}@cornell.edu, {maruf.zaber.09, mehrab.morshed, bahar61119}@gmail.com

ABSTRACT
The design of accessible mobile phone interfaces for low literate people usually assumes an individual model of use, and are often limited by the low technical expertise and/or cognitive ability of users in marginal communities of developing countries. Drawing on previous ICTD scholarship around shared and intermediated use of technology and our own ethnographic field study, we introduce a collaborative model of use in the design of *Suhrid*, a mobile phone interface that helps low literate users perform common phone tasks by receiving remote help from higher-literacy members of their community. The results of our six week long deployment of Suhrid among 10 low literate rickshaw pullers in Dhaka, Bangladesh, indicate the potential of collaborative use models to help low-literate people more effectively use mobile phones while strengthening bonds between them and the people in their community who provide help.

Categories and Subject Descriptors
H.5.2 [**User Interfaces**]: Interaction styles

General Terms
Human Factors

Keywords
Collaborative Interface; Low Literate User; Gift Economy

1. INTRODUCTION
Making technologies accessible to low-literate users is a long-standing challenge for ICTD researchers and practitioners. The rapid growth of mobile phone penetration in the developing world in the last two decades has driven important change across many aspects of life, from education and health to political participation and the informal economy. However, low literacy has limited these impacts, especially among poor and marginal populations; therefore helping low-literate users reap the benefits of mobile devices is an increasingly important question in ICTD research.

Previous work in this area has attempted to overcome this problem by using non-textual interfaces that incorporate graphic and audio based commands and content. While these efforts have shown benefit, they also face important limits, including the need for additional computational support from the device, which may not be available in low-resource environments, and the technical and cognitive abilities of users interacting with icons and audio commands. As a result, the design of an effective phone interface for low-literate people remains an ongoing challenge.

Our work is built around a shift in perspective about the use of 'personal' devices like mobile phones from an individual model to a more communal model, in which users figure not as atomized individuals, but as nodes within wider social networks that can be drawn on to overcome barriers that literacy poses to technology use. Prior studies in ICTD have shown that technology use in developing and low-resource contexts is frequently collective or distributed in nature, with the use of technologies like mobile phones shared among the user's family, friends, and other community members [4, 27]. Intermediate use of technologies is similarly prevalent; several studies have shown that low literate people often take help from digitally literate people close to them for operating their own mobile phones [23, 28].

At the center of intermediate use lies the practice of 'help' by able members of the community. We see an opportunity to leverage this practice through a community-sourced model that connects low-literate users to higher-literacy remote peers in their immediate network to both accomplish tasks and strengthen social bonds. In this paper, we present a design intervention that exploits the social values and practices of a community of rickshaw pullers in Dhaka and their intermediate use of technologies to provide low-literate members access to their basic mobile phone operations. Our previous ethnography on the same garage informed us of the intermediate use of mobile phones among this community. Based on that and an additional focus group study, we designed, developed, and deployed *Suhrid,* a phone application that allows the rickshaw pullers to remotely get help from their garage owner for placing phone calls and saving contacts. A six-week field deployment of Suhrid with 10 rickshaw pullers showed that it effectively helped low-literate users make better use of their phones. More generally, we argue that Suhrid shows the potential of designs that leverage shared and intermediate use in contexts where such use is common.

2. RELATED WORK
Literacy has long posed important challenges to mobile phone use among poor and marginal populations. In their 2006 paper, Chipchase et al. documented various ways illiterate people used mobile phones [7]. Their study showed that some functions (e.g., turning the phone on or off, accepting incoming calls) were easy for the users, while other functions (e.g., sending text messages, finding contacts from the contact list) were more difficult. Furthermore, understanding and responding to basic information

ACM DEV 2015, December 01 - 02, 2015, London, United Kingdom

ACM 978-1-4503-3490-7/15/11$15.00
DOI: http://dx.doi.org/10.1145/2830629.2830640

about the phone (e.g., remaining battery power, network connectivity, incoming text messages) was often challenging. These problems often led to confusion, mistakes, embarrassment, and non-use. A number of mobile handset vendors worked to address these problems by providing audio and visual clues for battery alerts, network connectivity, text messages, and so on [32]. However, problems related to operations like finding contacts from the address book and placing calls remain.

2.1 Designing for Individual Phone Use

A common design response to problems of literacy is to develop interfaces that use less text in order to reduce literacy requirements. For instance, icon-based interfaces have been used to support low-literate or illiterate populations of Indian village women [9], domestic laborers [20], and farmers [21], while audio interfaces have been developed for low-literate Pakistani health workers [30] and Indian farmers [25]. Color has also been used with some success to help low-literate users with address books [31] and phone contact lists [14]. A more detailed review of such design work can be found in Medhi [17].

These approaches face a number of practical challenges. For icon-based interfaces, removing text doesn't necessarily remove usability problems. Besides the difficulty of finding icons that make sense to low-literate users across a range of social and cultural contexts, hierarchical presentation of icons may fail due to cognitive challenges stemming from unfamiliarity or misrecognition of hierarchical orderings of information [18]. For audio interfaces, the users still need to remember the audio commands and their hierarchy. Such speech interfaces are also more error-prone than touch or graphical interfaces [25].

A common characteristic across each of the abovementioned design strategies was the assumption that a single user would use the device. This assumption leads to designs focused on communicating with that single user through graphical signals or audio clues; such designs depend on the individual user's memory and skills.

2.2 Communal Model of Technology Use

Individual use is not the only use model; a growing body of ICTD work has demonstrated that social use is common in low-resource contexts. For example, Burrell's ethnography in Ghana revealed how technologies like land-phones, computers, and televisions were shared among the members of a community [4]. Rangaswamy et al. found a similar sort of shared model of technology use in Mumbai slums [27, 28].

In the case of phones, the devices are not only shared between people, but also often used with the help of others, especially in the case of low-literate individuals. Parikh and Ghosh have discovered that low-literate Indian women take help from field workers to communicate with microcredit providers [23, 24]. Sambasivan et al. have reported the practice of helping in informal setting for operating mobile phones in India [29]. Kumar et al. have pointed to the network of actors beyond individual users supporting the consumption of mobile phone services in India [15]. Our previous work has shown how illiterate rickshaw-pullers in Dhaka, Bangladesh depend on literate social peers to support and access basic mobile phone operations [1].

This communal model extends to the systems of maintenance and repair by which devices and wider infrastructures are sustained in low-income environments. Work by Jackson et al. [12, 13] and Houston [11] with mobile phone repairers in Namibia, Bangladesh, and Uganda has shown the significance and extent of the local and global networks of materials and knowledge that sustain mobile phone use in many developing countries.

These studies connect in turn to ideas from Mauss' classic anthropological work *The Gift*, which has shown the centrality of networks and rituals of gift-giving and mutual support as a central feature of social life across a wide range of cultural contexts [16]. Drawing on Mauss, we argue that the essence of gift embedded in technical help produces and reciprocates honor, respect, and trust among the members of a community. This spirit is often hidden under the layers of official technical support infrastructures in the developed Western world, revealed only once those supports fall short [26]. Without such infrastructure, that spirit is more visible in developing countries through the shared and intermediate practices around technologies. Thus, asking for help in these contexts may be less of a burden or something to be avoided and more an integral part of community practice that allows people to provide gifts and thus strengthen bonds. Collectively, these sources have led us to think differently about the site and nature of mobile phone 'use', forming the starting point for the collaborative use model that underpins our design intervention.

In addition to collaborative use, we are also committed to collaborative design with specific target communities, both practically and ethically. A body of ICTD literature has pointed out the limitations of designing technologies in developing countries due to the lack of material infrastructures and resources [6], and the challenges of setting up new ones [10]. Here, involving the existing social infrastructure and community practices may transfer some responsibilities from technologies to the communities. Further, working closely with communities can lead to new design opportunities; Bellotti et al. have argued that designs that leverage the altruistic nature of human behavior should not only facilitate an extended access to different services through a shared economy model, but also create fields for newer technologies [2]. However, long-term engagement with local communities through ethnographic techniques is required to understand the social infrastructure, cultural values, and community practices. This sort of long-term engagement points to another central challenge of ICTD research, which has a long history of short term research projects that have often failed to respect (and thus, meet) the needs of specific contexts [8].

3. RESEARCH CONTEXT

Our current study was done with a community of rickshaw pullers in Dhaka, Bangladesh whom we studied through a six-month ethnography [1]. Here, we review key elements of the context from that study to help situate the current work.

We conducted our study in Kamrangirchar, a small area in Old Dhaka with many rickshaw garages. Usually the owners of the rickshaw garages buy the rickshaws and rent those to the rickshaw pullers on a half- or whole-day basis; the particular garage that we studied had 73 rickshaws, and Dhaka has a huge number of people (almost all male) who earn their living through rickshaw-pulling. This livelihood is not a wealthy one: after paying rent to the garage owner the daily income of a rickshaw puller in a day ranged from 300 Taka ($3.80[1]) to 800 Taka ($10), and this is in the lower tier of the society.

[1] 1 Taka is equivalent to approximately 1.2 US cents.

On average, the rickshaw pullers are also in a lower educational range in society. Most of the rickshaw pullers dropped out of primary school, while the rest never went to a school; often this was because they had to help support their families [22]. None of them could read or write a complete sentence in any language. They were familiar with Bangla and English digits, but they could not read numbers when two or more digits were put together. Thus, these rickshaw drivers are a largely low literate population.

Still, the use of mobile phones is very common among the rickshaw pullers, mirroring general trends of increased phone use among lower income residents of Dhaka over the last five years [5]. Our previous study revealed that all of the rickshaw pullers had their own mobile phones. Most of those mobile phones were Java-enabled China-made devices costing between 2,000 Taka ($25) to 10,000 Taka ($120). Many of them bought their mobile phones from second-hand markets or mobile repair shops at a savings of about the 50% of the original prices. The rickshaw pullers reported that they would spend around 20-30 Taka (25-35 cents) per day using their phone.

Basic functions such as making calls and saving contacts were not easy for the rickshaw pullers because reading and writing contact names and numbers requires literacy skills, and although they tried workarounds described in the design goals section below, these often led to frustration. In practice, they would often get help from their garage owner. Although the garage owner was not educated in formal schools, he had basic literacy skills, and as a tech-savvy person, owned several models of smartphones and had used a variety of other phones. His literacy and technical knowledge, combined with the social and economic connections between him and the pullers, led the garage owner to be a primary support for the pullers in using their phones [1]. He also arranged for electric power supply, outlets, and phone adapters in the garage so that the rickshaw pullers could recharge their phones while resting.

We chose this particular rickshaw puller community because of the fit with our condition of low-literacy, the use of phones, the social structure of help giving, and our convenience to reach and study them. All the members of our team who worked in the field were affiliated with a local university close to the garage. All of them were born and brought up in Bangladesh, and speak Bangla like the rickshaw pullers.

4. *Suhrid*

We turn now to the design of Suhrid (Bangla for ‘a good friend’), an application to support the low-literate rickshaw pullers in getting help from their social connections for two basic operations on their mobile phones: 1) placing a phone call to a contact and 2) saving a contact. We first discuss the major design goals and constraints that arose from prior work and our own interactions with the garage owner and pullers, then how the design and implementation of Suhrid supports them.

4.1 Focus Group and Design Goals

We conducted four focus group discussions at the garage in the evening, when the rickshaw pullers were usually back from their work. The number and identity of participants were not the same in each discussion; rickshaw pullers who were present in the garage at those times and were interested in discussing joined, leaving in the middle if they had work to do. However, there were almost always at least six pullers present in these discussions, along with the garage owner. We discussed with the rickshaw

Figure 1. A focus group discussion at the rickshaw garage. The garage owner (wearing a cap) participated with seven other rickshaw pullers in that session. (The picture is taken and shared with proper permission of the people in the picture. The faces are blurred for anonymity)

pullers questions around the length, purpose, and patterns of their mobile phone use. Participants were not paid for their participation in this round of the study; however, food and drink were supplied during our discussions. The discussions were audio recorded with their permission, and later transcribed in Bangla and translated into English. Two members of our team coded the discussions independently; their findings were matched in a group discussion among the team and the main themes were extracted.

All of our rickshaw puller participants reported that the two main tasks that they did with their mobile phones were placing and receiving phone calls. They shared their difficulties in finding a contact from their contact list. They said they often tried to memorize the contact numbers by face, sometimes they tried to remember the position of a contact in their list, and so on. These processes often did not work well, and they ended up choosing the wrong person to call. Then they became embarrassed after talking to the wrong receiver, and sometimes they decided not to place a phone call to avoid this embarrassment. They also wasted money when they placed phone calls to wrong numbers.

The rickshaw pullers also reported that they often struggled to find ways to save a new contact on their phone. Sometimes they used punctuation symbols to remember the contact associated with a given number (e.g., “# is for *Mr. Choudhuri*; ## is for *Mr. Mitra*”), but then later forgot. They also found it difficult to save a number from their dialed, received, or missed call lists. They said they usually took help from their garage owner for these tasks. The garage owner also agreed that he often helped them for these. However, they informed us how they struggled when they were away from their garage. All the participants opined that a mobile phone application that could help the rickshaw pullers to remotely get that help from the garage owner would help them.

An immediate design challenge was the functionality of the mobile phones that our participants were using. Since most of them were not using smartphones, and they used a variety of devices, it was difficult for us to design a single interface. However, one of the rickshaw pullers suggested that many of them were thinking to buy smartphones soon. The rest agreed, sharing their experiences of using smartphones owned by friends or relatives. We shared that designing on a smartphone would be easier for us. We showed our own smartphones to them, and briefly gave them an idea about the interface. They all expressed

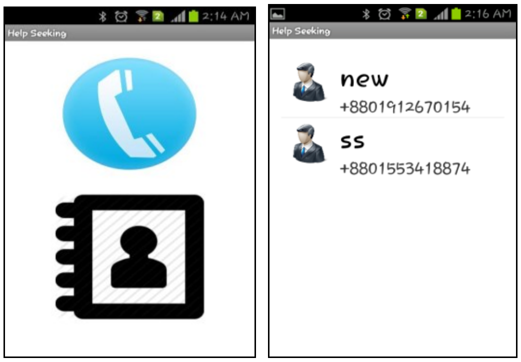

Figure 2. The first version of UI for low-literate rickshaw pullers. (Left) The interface for selecting if they wanted to place a call or save a contact. (Right) The list of helpers. In both cases the goal was to minimize the number of elements on the screen and the need for literacy skills.

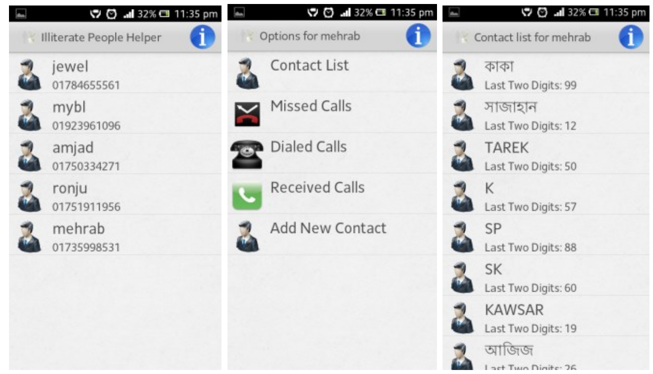

Figure 3. Three screenshots of the helper side interface. On the left, the list of seekers subscribed to this helper. In the middle, the contact list, lists of missed, dialed, and received calls, and an option to add a new contact to the contact list. On the right, the contact list of one of the seekers, showing only the last two digits of the phone number to address privacy concerns.

their excitement about having an application for them in a smartphone. We asked them if smartphones would be affordable for them. They showed us the costs of some of the Java enabled phones that they were using, and some cheap Chinese-branded smartphones in the market, and argued that the differences would be minimal. They further argued that the better quality of the pictures and video on a smartphone would rationalize the additional cost they would bear. Based on their recommendation, we decided to design an Android phone application for helping the rickshaw pullers reach their helpers when they were away from them.

4.2 Interfaces

Suhrid has two interfaces: one for the low-literate users ("seekers") and one for the help providers ("helpers"). Based on our findings, we emphasized the two basic phone operations that the rickshaw pullers needed the most: placing calls and saving contacts. Both our own work and previous studies show that low-literate people often get confused with too many icons in the interface [18]. Hence, on the seeker's side, Suhrid starts with an interface with only two icons (Figure 2, left). The top icon was for sending a request to place a call and the bottom icon was for sending a request to save a contact. The rickshaw pullers chose the icons during our group discussions. After choosing one of these two icons, the next screen appears with the list of helpers (Figure 2, right). We used comic fonts in that interface to give that an informal and friendly look. Touching a helper's name would place a call from the seeker's phone to that helper's phone through Suhrid.

On the helper's side, Suhrid displayed the list of seekers subscribed to that helper (Figure 3, left). On choosing any seeker, Suhrid would show a list with five entries: a) contact list, b) missed calls, c) dialed numbers, d) received calls, and e) an option to add a new contact to the contact list (Figure 3, middle). If the helper chose any of these, he could see the corresponding list. However, our previous study showed that some rickshaw pullers were concerned about the privacy of their contacts [1]. It also reported that the rickshaw pullers would often remember the contacts by the last three digits of their phone numbers. So, we left the last two digits visible to facilitate referencing the correct phone numbers between the seekers and the helpers, but hid the rest of the number to increase the pullers' confidence that the numbers would be private (Figure 3, right).

4.3 Functions

4.3.1 Seeking help

We found in our field study that rickshaw pullers would only get help from the people they knew. The garage owner was also interested in helping the rickshaw pullers because he knew them. Hence, we chose a system that paired people who knew each other (versus an anonymous crowdsourcing model). In Suhrid, the pairing up could be done by adding a helper on the seeker's interface and adding a seeker in the helper's interface. Since this task needed some competency with the mobile phone, we assumed that the helper would do this task when they were co-present with a seeker they agreed to pair up with.

For seeking help, the low-literate user needed to open Suhrid on their mobile phone, select whether to place a call or save a contact, and choose the helper from the list. If it was the first time the seeker was calling the helper through this application, then the list of incoming, outgoing, and missed calls, and the contact list would be sent from the seeker's phone to the helper's phone. If it was not the first time, then the system only sent changes of these lists since the last time they contacted that helper.

4.3.2 Placing calls

If the seeker requested help for placing a call to somebody from their contact list or from dialed, received, or missed call logs, then the helper could easily find that on his interface and find the appropriate name. The helper would choose that contact and the application would send that number to the seeker's phone. The seeker's application would receive the number and place a phone call from his phone after getting a confirmation from the seeker. This way, the low-literate people could place a phone call to anybody they wanted.

4.3.3 Saving contacts

If the seeker asked help for saving a contact to their contact list that could also be done in a similar way. The seeker could request to save a number from their dialed, missed, or received calls' lists, or could just tell the helper a phone number to save with a particular name. The helper could find the number in one of the lists, or just simply type it on the software screen. After getting the number, the helper could save the contact with the appropriate name into their local copy of the helper's contact list. Suhrid would then send a message from the helper's phone to the seeker's, which the software would intercept and use to update the contact list on the seeker's phone. This way, the new contact would be saved in both phones.

4.4 Implementation

The application was built for Android-based smartphones using Java. To avoid the need for an Internet connection, communication between the mobile phones was done through text messages. For example, when the seeker's phone needed to send the contact list to the helper's phone, the application on the seeker's phone would compose a formatted text message putting the contacts' names and mobile phone numbers one after another. This message was often long, and the system needed to break the message down to several text messages for sending. On the helper's phone, the application would receive these messages one after another, and re-construct one single message. Then the application would parse that message to extract each contact and make a contact list for the seeker in the helper's phone.

5. USABILITY EVALUATION

After developing Suhrid, we conducted one laboratory study and two field level user studies to understand the usability of this application. We made necessary changes to Suhrid based on our findings in these studies.

5.1 Laboratory Study

We invited the rickshaw pullers and the garage owner to visit our university in order to introduce them to our system formally and to see if they had any difficulties in using it in the lab. The session was two hours long, and each of them was paid 800 Taka ($10), which was equivalent to a puller's earnings for their whole day and also satisfied the garage owner. There were three reasons behind this session. First, we had enough smartphones in our laboratory so everyone could use one. Second, they were interested to see the laboratory and how we work. Third, we believed that this session helped us develop a better relationship with our participants.

A total of 12 rickshaw pullers and the garage owner came to visit our laboratory. We also invited four undergraduate students to help in this session. The demonstration session was conducted in Bangla and lasted for three hours. We first lectured them about the use of a smart phone and then we showed them its basic features. To provide them first-hand experiences we divided our participants into four groups, each consisting of three rickshaw pullers and one undergraduate. Each group was given a smartphone to play with and the undergraduate, who was an expert smart phone user, helped them.

First we briefed them how to use the "touch" action to operate the smartphones. Participants picked up the basic operations (touching instead of pressing buttons, turning a phone on or off, etc.) very quickly. We then demonstrated Suhrid. After that interactive session, the participants were allowed to practice the use of our application. To mimic the real-life scenario, their helper, i.e., the garage owner, was taken to another room so that he could not verbally communicate with the participants, but could communicate with them through our application.

All 12 rickshaw puller participants were told to ask the garage owner through Suhrid for help in placing a phone call and saving a contact. Five of them could perform both of the tasks in the first attempt without any help. The average time they took to place a call was 40 seconds, and the average time for saving a contact was 1 minute. Four of them forgot the process, so our team members helped them remember. Three of these four participants could complete both of these tasks after the reminder. The remaining person needed instruction one more time. The other three participants performed the calling task twice; apparently they misunderstood our instructions. However, when we explained the task again, they could perform both tasks in the first attempt.

After the tasks, we asked them about their general impression about Suhrid. All of them expressed their excitement around it. They said the reason why some of them had initial difficulties was because both smartphones and the application were new to them. We then asked the garage owner about his experience. He said that he enjoyed the whole process, and praised the software.

5.2 Field Level User Study, First Round

Although the rickshaw pullers and the garage owner performed satisfactorily in the lab, and expressed their satisfaction in using the application, we wanted to understand if that would be reflected in the context they spend most of their time. So, the next week, we conducted a field level user study at their garage with 10 of the 12 participants who attended our demonstration session. We tried to replicate a similar situation of help-seeking there. We made sure that the garage owner/helper couldn't communicate verbally with the participants and that other rickshaw pullers present in the garage did not help them either. As before, we paid each participant the average income of a whole day.

5.2.1 Call Generation

Each participant was asked to generate three phone calls to numbers saved in their phonebook. Three of them hesitated in the beginning as they thought they might break the expensive smart phones by their inexperienced use. However, when we assured them that no such thing would happen, they started using the smartphone. Five rickshaw pullers succeeded in all three trials. The other five made mistakes in the first trial, but succeeded in the second two trials.

The first trial of call generation took on average 45-50 seconds including the time to press the call button, select the helper, talk to the helper over the voice call, and receive the response from the helper's application. Both the participants and we considered this time much longer than the usual time one takes to place a phone call to somebody. According to our observations and their feedback, inexperience with touchscreens, lack of confidence, and confusion of the "save" icon with the "call" icon were the main reasons for this delay. The average time for successful call generation was eventually reduced to 30-35 seconds in the next two trials, which the rickshaw pullers considered good enough.

5.2.2 Contact Saving

Next, we asked each of our participants to save a random contact number or any unsaved contact number from their call history using Suhrid. Like before each of them attempted the test thrice. This time, their confidence using the touch screen seemed to have improved. However, only three of them were successful in all three attempts. Two participants failed in all three attempts. After the test they explained that the low success rate in contrast to the call test was mainly due to their confusion with the two icons in the interface, struggling with differentiating the functions of two graphical objects, one for 'calling' and the other for 'saving a contact'. This observation matches with Medhi et al.'s claim about the weakness of low-literate people in handling the cognitive load associated with graphics [17, 19].

5.3 Field Level User Study, Second Round

In our next round of design, we responded to these problems by removing the initial interface for choosing a service on the seeker's UI. Upon opening the app, the list of helpers would appear immediately (Figure 2, right). On selecting the helper, a call would be generated to the selected helper. The seeker would then tell the helper whether he was trying to call somebody or save a contact. The helper then would use the appropriate lists through Suhrid (Figure 3, middle) to find the number. Upon selecting the number, Suhrid would offer the helper an option for either making a phone call or saving that number in the contact list. We made the necessary changes in the application according to this design, and then conducted another round of field level user study at the rickshaw garage in Kamrangirchar two weeks after the first round.

We used the same 10 participants, setup, payment, and task that we did in the first round. Eight out of 10 participants were successful in all three attempts on both tasks this time. The other two failed in in one attempt each. One of them ran out of credit in the middle of the process while the other one asked the helper to help him call to a number that had not actually been saved in his phonebook. In one case the helper mistakenly saved a different number from the call log, so we repeated that test and the rickshaw puller succeeded. The overall remarks of the participants were overwhelmingly positive. One participant commented,

"*Just one click and some other person does the job from another place! I don't have to do anything!! It will be a great help for me if I use this phone. Not only for me, but also for every low literate and illiterate people will be happy to use this application*."

6. DEPLOYMENT

After the success in the second round of field usability testing, we decided Suhrid was ready for a real deployment to understand better how Suhrid would help this rickshaw puller community. The rickshaw puller participants were chosen based on their interest to participate in our study, which lasted six weeks. Ten smartphones were given to the pullers and two other phones were given to the garage owner and his brother who would work as the helpers. All participants kept the smartphone as compensation. The garage owner and the rickshaw pullers recommended the owner's brother as a second helper. He would often come to the garage and help them in operating their mobile phones the same way the garage owner did, and was willing to participate in the deployment and able to operate Suhrid. So, in the helpers' list on Suhrid, each of the rickshaw pullers had the number of the garage owner at the top followed by the number of his brother.

The lists also had a third number. We were curious about whether people would take help from strangers when their normal helpers weren't available, in the way that some crowdsourcing applications such as *VizWiz* [3] enable people to get help from strangers. To do this, we recruited two freelance workers, students of a local university from outside the rickshaw puller community. After talking to both the rickshaw pullers and the freelance workers, it was decided that the rickshaw pullers would pay 2 Taka (3 cents) each time they asked the freelancers for help. The rickshaw pullers were clearly informed that they would have to pay when they would seek help from these freelancers. Each freelancer was assigned to be the third and final helper for half of the rickshaw pullers.

To evaluate Suhrid, we collected three main sources of data. First, we collected usage data through a text message Suhrid sent each time a rickshaw puller asked for help that included the seeker, helper, and the time. We received permission from our participants to collect these data. Second, at the end of six weeks, we went to the garage and interviewed each of our participants about their experiences throughout the deployment period. These interviews were audio recorded with the participant's permission and conducted in Bangla. The average length of the interviews was 30 minutes and participants were given 200 Taka ($2.5) each as compensation. Finally, we invited the rickshaw pullers to our university guest room one more time. This was both a focus group discussion and a formal conclusion of the study. All of the participants were paid 800 Taka ($10) for this two-hour session. We did not invite the garage owner and his friend this time to allow the rickshaw pullers to express their experiences freely. Instead, we had a separate group discussion with the garage owner and his brother at the garage.

In both of the focus group discussions we asked them about their experiences of using Suhrid. Those questions included if they had any difficulties in using Suhrid, in which situations they used Suhrid, if they got help whenever they needed, what their experiences were around the freelance workers, why their usage of Suhrid declined over time, and if they had any other issues with Suhrid. The transcription, translation, coding, and resolution processes were the same in this round as before.

6.1 Results

The interviews and focus group discussions revealed a number of important aspects around the use of Suhrid. Participants described the ease of using the software, its value and availability, and the appreciation it gave them for the giving of help between them. They also expressed concerns about privacy and reluctance to take help from the freelance workers, as well as the gradual decline of their usage of Suhrid over time.

6.1.1 Ease

All seekers were satisfied with the ease of using Suhrid. One of them said,

"*This software is very easy to use. I didn't find any difficulty while using it. See? Here is the icon I had to press to call the Mahajan. And if the Mahajan is not available I press the second number*." [Mahajan means the 'The big person'. Here he referred to the garage owner. The second number was his brother's.]

The helpers also found Suhrid easy to use. The garage owner said,

"*When you first gave me the software I understood the full functionality. Here is the list and I can select any of them and see the list of the missed call, received call and their phone book. .... It's very easy to use.*"

6.1.2 Availability

Suhrid also provided the seekers valuable services and help, sometimes at critical times. As one participant said,

"*I was at the hospital for a checkup of my father and I needed to call a doctor. Another person saved the number, I remembered the name by which he saved it but couldn't find it. It was an emergency. I called the Mahajan and asked him to dial the number for me. He did and I talked with the doctor*."

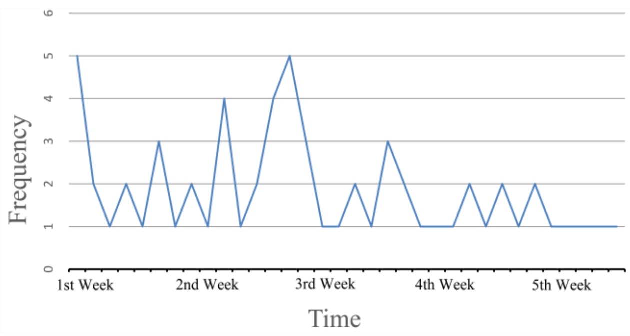

Figure 4. Daily usage of Suhrid during field deployment. Use of Suhrid was higher in the first few weeks, then declined.

The seekers shared how Suhrid helped them by making the helper available in the contexts they needed. One participant said,

"*Normally when I get out of home to go to Mahajan I forget by the time for which reason I went there. So, most of the time that doesn't help me. But with this software I can easily call him and ask him to save a contact for me. Sometimes it takes time but work gets done.*"

Seekers almost always got help from their helpers, first trying to get help from their garage owner, then the owner's brother. In only five cases, they could not get help from either of them. In three such cases, they took help from the freelance workers. In the remaining two cases, they stopped trying.

6.1.3 Reluctance to Take Help from Freelancers

The seekers expressed their lack of interest in receiving help from the freelance workers. All of them said that they did not need to call the freelancers because they had almost always got help from their garage owner. Also, they reiterated that help should come from people they knew. One of them said, "*I don't feel comfortable asking help to somebody I do not know*." Once he said that, the rest of the participants agreed, saying, "*yes*". One seeker who took help from one of our freelance workers said,

"Although I took help from him, I never asked him how he was doing. It was only regarding help. I only called him for help nothing else. He was always my last option for help."

6.1.4 Improving Communal Bonds

In contrast to their experience with freelancers, both the seekers and the helpers reported that using Suhrid with people they knew strengthened their community feelings. The concentration of help requests on the garage owner and the brother was not a concern to them; both reported that they had enjoyed their task. We asked them if the number of incoming helping requests ever bothered them. The garage owner said that he always had his mobile phone with him, and the time and efforts it needed to help one was not overwhelming to him. However, he also mentioned that he could not respond to couple of calls as he was busy in saying prayers.

All of the rickshaw pullers reported that the availability of their garage owner all the time helped them realize how much care the garage owner had for them. One rickshaw puller said, "*Now I realize how caring he (the garage owner) is. You cannot help somebody like this if you do not care for them a lot.*" The garage owner said that the relationship between him and the rickshaw pullers improved because of Suhrid because now they were taking help even after leaving the garage. He said, *"It is great to come closer, and to make them understand how a well-wisher should be. Now they know better how much I care for them. You can see the added respect in their words these days."*

6.1.5 Gradual Decline of Usage

Despite the value people drew from using Suhrid, its use declined over time. In total, it was used 63 times over the span of six weeks (Figure 4). The number of requests sought for placing a call was much higher (41) than for saving a number (22). Among the 63 requests, the garage owner received 46, his brother received 14, and the freelancers received 3.

Participants informed us that usage declined because initially they had to take more help to save the new contacts, so the number of requests was high. Also, they often call the same numbers, which they learned to call themselves after taking help once or twice. Then they would only take help whenever they had to call a new number or a save a new contact. Furthermore, they still often took help from the garage owner in person when both were in the garage together; Suhrid was mainly useful when the puller and owner were not in the same place.

6.1.6 Concerns

The seekers also shared several concerns around Suhrid. One of them complained that our software damaged his mobile balance. He said,

"I had 50 taka in my mobile when I called Mahajan but when I finished talking to him, I noticed after some time that my balance is zero. I was confused." (500 Taka is approx. $6)

Later, we discovered he had not turned off his mobile phone after calling, and that caused a big deduction from his balance. Another seeker wondered about the privacy of the contacts saying,

"*I didn't understand how the garage owner was able to call my wife through my phone. And how did he save contact numbers in my phone remotely? Was the number saved in his phone too?*"

We explained to him why it would not be possible for the garage owner either to know the phone number or to place a call to his wife. He seemed to be satisfied with our explanation.

7. DISCUSSION

The design, development, and deployment of Suhrid generate a number of immediate lessons for better designing a UI for low-literate people, and some larger lessons pertinent to problems of design in the context of developing countries.

First, our study supports previous findings around the struggle of low-literate users with hierarchical presentation of information on graphical interfaces [18]. The confusion that arose in the first field study with the selection interface went away with the removal of that screen, suggesting the value of minimizing hierarchy in the interface. The fact that the rickshaw pullers struggled with even two icons on the selection screen, but were able to sequentially navigate through the list of three helpers, may indicate that low-literate users might be better at memorizing the relative positions of contacts than the interpretation of symbolic icons.

Second, our study suggests that privacy management is important in designing shared-use interfaces. Contact information can be confidential and the privacy associated with different parts of contact information is dependent on the users' interpretation. The fact that rickshaw pullers used the last three digits of phone numbers to identify contacts had to be respected in our design:

showing only two digits balanced the needs for communicating between helpers and seekers while respecting seekers' perceptions of privacy. A related sensitivity concerned asking for help from people outside the community. We found our participants preferred to get help from another person in their own community over somebody from outside. This particular finding may suggest limits to more generic crowd-sourced solutions, and encourage future researchers to weigh local sensitivities before advancing crowd-based responses to local use challenges in such contexts.

Third, like many other technologies for developing countries, Suhrid demonstrates how cost influences design choices. The community, according to their cost-benefit analysis, rationalized the choices of the Android platform and text-based communication. The credit in the balance that was spent for a typical use of Suhrid, including both the call itself and the text messages sent in the background, was around 70 to 90 paisa, or about one US cent, for the seeker and 30 to 40 paisa for the helper. Although we reimbursed participants in the end for all the text messages Suhrid generated during the deployment, in general participants would need to pay for Suhrid themselves. Thus, we asked our participants if this expenditure would be heavy for them. They all considered this expenditure justified for the service. One participant said,

"*A rickshaw puller is not as poor as you think. I usually spend 500-600 Taka for my mobile phone every month. And I would be happy to spend 1-2 Taka if I could get my purpose served with remote collaboration.*" (500-600 Taka is approx. \$6-\$7, and 1-2 Taka is approx. 2 to 3 cents)

Fourth, we observed that the low-literate rickshaw pullers started learning the contacts while taking help from Suhrid. While explaining the gradual decline of the usage, they mentioned how they could recognize the old contacts from their memory and they only needed help for the new contacts. They said they could recognize the faces of the contact names or numbers. This finding suggests that such a help system might also eventually educate and empower individual users over time. This gradual decline in help-seeking also suggests that their cost for using Suhrid would likely decline over time as well.

Beyond the lessons for immediate design, our study indicates a number of bigger concerns for ICTD. First, our study showed that extending the reach of help strengthened the relationship between the rickshaw pullers and the garage owner. The garage owner expressed his satisfaction to be able to come closer to the rickshaw pullers because of Suhrid. Likewise, the rickshaw pullers praised the garage owner for helping them in important times through Suhrid. Thus, beyond its immediately instrumental effects in extending effective use among low-literate populations, Suhrid performed a secondary but no less important role in strengthening and reinforcing local social relations, notably relations of respect and trust between the garage owner and rickshaw pullers. This finding echoes in modest form Mauss' classic finding around the importance of mutual aid and gift giving as a central and indeed constitutive moment of social life [16]. If the goal of ICTD work is to empower and support not only users but also the wider networks and communities of which they are a part, design options that leverage and extend core features and principles of sociality itself ought to figure more prominently in the field.

Second, our study demonstrates the importance of anchoring design interventions in longer-term programs of ethnography and community engagement. Unlike emphasizing scaling up 'one size fits all' technologies, we ground our designs in the values, norms, and practices of a particular community through our engaged field work. This helped us address many nuances of design that would be otherwise difficult to find out. For example, decisions like using the Android platform, remote help, revealing only two digits of phone numbers, or communicating through text messages came out of the users' values and practice.

Third, our deep engagement with the community helped relax the power differences between the designers and users, and opened up opportunities for designing technologies with their participation. The rickshaw pullers often discussed with us other problems of their life, along with a range of ideas towards solutions. For example, some rickshaw pullers joined a co-operative society where they needed to pay regularly, and they needed help to determine and remember saving from their everyday income. We are now working with them towards technology solutions that may help with this goal. While such long-term and locally responsive models of community engagement may limit the 'scaling up' of single technologies to wider social and cultural contexts, they suggest new opportunities to design multiple, appropriate technologies for the same community—opportunities that may extend the depth, impact, and responsibility of ICTD work. They also demonstrate the kind of long-term commitment to places and individuals that may serve as a partial antidote to the kinds of "research tourism" [8] long identified and criticized in ICTD work.

8. CONCLUSION

In this paper, we have described the design, development and deployment of our community sourced mobile phone interface for a low literate rickshaw puller community in Dhaka. Building on our ongoing ethnographic work, Suhrid seeks to leverage and extend existing distributed practices of technology use to overcome literacy-based barriers to mobile phone use within our target population. We involved the members of the community in each step of our design process, and conducted two rounds of field level user studies to refine our design. Finally, we deployed Suhrid for 6 weeks and conducted a post-deployment user study through interviews and focus group discussions to understand its successes and limitations. Our study generates a number of lessons the use of icons, privacy, and cost-benefit negotiation in designing such collaborative interfaces in developing countries. Furthermore, we present our arguments supporting the potential of gift-based design and long-term engagement in ICTD works.

9. ACKNOWLEDGEMENT

We thank the garage owner, and the rickshaw drivers, and the freelance workers for their support during this study. We thank the Department of Computer Science and Engineering (CSE) of Bangladesh University of Engineering and Technology (BUET) for providing us laboratory facilities. We also thank International Fulbright Science and Technology Fellowship and Intel Science and Technology Center for Social Computing (ISTC*SC) for supporting this research.

10. REFERENCES

[1] Ahmed, S.I., Jackson, S.J., Zaber, M., Morshed, M.B., Ismail, M.H.B. and Afrose, S. 2013. Ecologies of Use and Design: Individual and Social Uses of Mobile Phones Within Low-Literate Rickshaw-Puller Communities in Urban Bangladesh. In *Proc. DEV-4*, 14:1–14:10.

[2] Bellotti, VM. et al. 2014. Towards community-centered support for peer-to-peer service exchange: rethinking the timebanking metaphor. In *Proc. CHI'14*, ACM, 2975–2984.
[3] Bigham, JP. et al. 2010. VizWiz: nearly real-time answers to visual questions. In *Proc. UIST'10*, ACM, 333–342.
[4] Burrell, J. 2010. Evaluating Shared Access: social equality and the circulation of mobile phones in rural Uganda. *Journal of Computer Mediated Communication*. 15, 2 (2010), 230–250.
[5] Chen, G. and Rasmussen, S. 2014. *bKash Bangladesh : A Fast Start for Mobile Financial Services*. World Bank.
[6] Chen, J. 2015. Computing within Limits and ICTD. In *Proc. LIMITS'15*.
[7] Chipchase, J. 2006. How do you manage your contacts if you can't read or write? *Interactions*. 13, 6 (2006), 16–17.
[8] Dearden, A. 2013. See no evil?: ethics in an interventionist ICTD. *Information Technologies & International Development* 9, 2, 1-17.
[9] Ghosh, K., Parikh, T.S. and Chavan, A.L. 2003. Design considerations for a financial management system for rural, semi-literate users. In *Proc CHI EA'03*, ACM. 824–825.
[10] Heimerl, K., Hasan, S., Ali, K., Parikh, T. and Brewer, E. 2015. A Longitudinal Study of Local, Sustainable, Small-Scale Cellular Networks. *Information Technologies & International Development*. 11, 1, 1-19.
[11] Houston, L. 2014. Inventive Infrastructure: An Exploration of Mobile Phone. PhD Thesis, Lancaster University.
[12] Jackson, S.J., Ahmed, S.I. and Rifat, M.R. 2014. Learning, innovation, and sustainability among mobile phone repairers in Dhaka, Bangladesh. In *Proc. DIS'14*, 905–914.
[13] Jackson, S.J., Pompe, A. and Krieshok, G. 2012. Things fall apart: maintenance, repair, and technology for education initiatives in rural Namibia. In *Proc. iConference'11*, 107–116.
[14] Joshi, A., Welankar, N., Kanitkar, K. and Sheikh, R. 2008. Rangoli: a visual phonebook for low-literate users. In *Proceedings of the 10th International conference on Human computer interaction with mobile devices and services*, 217–223.
[15] Kumar, N. and Parikh, T. 2013. Mobiles, music, and materiality. In *Proc. CHI'13*, 2863–2872.
[16] Mauss, M. 1954. *The gift: Forms and functions of exchange in archaic societies*. IG Cunnison. London: Cohen and West.
[17] Medhi, I. 2015. User Interface Design for Low-Literate and Novice Users: Past, Present and Future. *Foundations and Trends in Human-Computer Interaction*. 8, 1, 1-72.
[18] Medhi, I., Lakhmanan, M., Toyama, K. and Cutrell, E. 2013. Some evidence for the impact of limited education on hierarchical user interface navigation. In *Proc. CHI'13*, 2813–2822.
[19] Medhi, I., Prasad, A. and Toyama, K. 2007. Optimal audio-visual representations for illiterate users of computers. In *Proc. WWW'07*, 873–882.
[20] Medhi, I., Sagar, A. and Toyama, K. 2006. Text-free user interfaces for illiterate and semi-literate users. In *Proc. ICTD'06*, 72–82.
[21] Medhi-Thies, I., Ferreira, P., Gupta, N., O'Neill, J. and Cutrell, E. 2015. KrishiPustak: A Social Networking System for Low-Literate Farmers. In *Proc. CSCW'15*, 1670–1681.
[22] Meheri, T. 2012. *Rickshaw Cycle Drivers in Dhaka: Assessing Working Conditions and Livelihoods*. Erasmus University.
[23] Parikh, T.S. and Ghosh, K. 2006. Understanding and designing for intermediated information tasks in India. *IEEE Pervasive Computing*. 5, 2, 32–39.
[24] Parikh, T.S., Ghosh, K. and Chavan, A.L. 2003. Design studies for a financial management system for micro-credit groups in rural India. In *ACM SIGCAPH Computers and the Physically Handicapped*, 15–22.
[25] Patel, N., Chittamuru, D., Jain, A., Dave, P. and Parikh, T. 2010. Avaaj Otalo: a field study of an interactive voice forum for small farmers in rural India. In Proc. CHI'10, 733–742.
[26] Poole, E.S., Chetty, M., Morgan, T., Grinter, R.E. and Edwards, W.K. 2009. Computer help at home: methods and motivations for informal technical support. In *Proc CHI'09*, 739–748.
[27] Rangaswamy, N. and Nair, S. 2010. The mobile phone store ecology in a Mumbai slum community: Hybrid networks for enterprise. *Information Technologies and International Development*. 2, 3, 51-65.
[28] Rangaswamy, N. and Sambasivan, N. 2011. Cutting Chai, Jugaad, and Here Pheri: towards UbiComp for a global community. *Personal and Ubiquitous Computing*. 15, 6, 553–564.
[29] Sambasivan, N., Cutrell, E., Toyama, K. and Nardi, B. 2010. Intermediated technology use in developing communities. In *Proc. CHI'10*, 2583–2592.
[30] Sherwani, J., Ali, N., Mirza, S., Fatma, A., Memon, Y., Karim, M., Tongia, R. and Rosenfeld, R. 2007. HealthLine: Speech-based access to health information by low-literate users. In *Proc. ICTD'07*, 1–9.
[31] Wiedenbeck, S. 1999. The use of icons and labels in an end user application program: an empirical study of learning and retention. *Behaviour & Information Technology*. 18, 2, 68–82.
[32] *Global Accessibility Reporting Initiative (GARI): Mobile Handset/Device Accessibility Report.* Available at: http://www.gari.info/